

(1)

Differential Equations

Lecture 01 :

25th August, 2026.

Differential Equation:

Definition :

An equation which involves differential coefficient or derivatives of dependent variables with respect to independent variable is called "differential equation".

(1) $\frac{dy}{dx} + y \cos x = \sin x$

(2) $\frac{d^2y}{dx^2} + xy \left(\frac{dy}{dx} \right)^2 = 0$

(3) $\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{3/2} = \frac{d^2y}{dx^2}$

(4) $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = nx$

(5) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$

Types Of Differential Equation :

There are two types of differential equation.

- ①- Ordinary differential Equation (O.D.E)
- ②- Partial Differential Equation (P.D.E).

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Ordinary Differential Equation :

The differential equation which involve a single independent variable are called ordinary differential equation.

e.g =

$$(1) - \frac{dy}{dx} = x + y$$

$$(2) - \frac{dy}{dx} + y \cos x = \sin x$$

$$(3) - \frac{d^2y}{dx^2} + xy \left(\frac{dy^2}{dx} \right) = 0 -$$

Partial Differential Equation :

The differential equation which involve a partial differentiation coefficients w.r.t more than one independent variable is called Partial differential Equation.

e.g =

$$u = f(x, y, z)$$

$$\bullet \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0 -$$

(3)

Order and Degree Of Differential Equations:

Order of Differential Equation:

The order of a differential equation is the order of the highest differential coefficient which occurs in it.

e.g.:

① - $\frac{d^2y}{dx^2} + \frac{dy}{dx} + 5x = 0$. (order is 2)

② - $\frac{d^4y}{dx^4} + \sin x = 5$ (order is 4)

③ - $y''' + 3xy'' + 5y' = 6$ (order is 3)

Degree of differential Equation:

The degree of a differential equation is defined if it can be written as a polynomial equation in the derivatives and for such differential equation its degree is given by the highest power of the highest order derivative appearing in it. Provided the derivatives are made free from radicals and fractions.

*- degree of differential equation is always a positive integer.

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* Degree $\neq -ve$
 * No fraction applicable

3rd September 2024.

$$(1) - \left(\frac{dy}{dx} \right)' + y = 0$$

order 1, degree 1

$$(2) - \left(\frac{d^2y}{dx^2} \right)' + \frac{dy}{dx} + 5 = 0 = \text{order } 2, \text{ degree } 1$$

$$(3) - \left(\frac{d^2y}{dx^2} \right)^{\frac{3}{2}} + \left(\frac{dy}{dx} \right)^4 + 5 = 0 = \text{order } 2, \text{ degree } 3.$$

* highest order ki power ko dekhengey wohi degree hogi.

→ big derivative is in angle

$$\frac{d^4y}{dx^4} + \sin(y''') = 0 = \text{order } 4, \text{ degree not defined}$$

$$\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{3/2} = 5 \frac{d^2y}{dx^2}$$

Squaring On Both Sides:

$$\left[\left\{ 1 + \left(\frac{dy}{dx} \right)^2 \right\}^{3/2} \right]^2 = \left\{ 5 \frac{d^2y}{dx^2} \right\}^2$$

$$\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^3 = 25 \left(\frac{d^2y}{dx^2} \right)^2$$

order = 2, degree = 2.

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'Lecture 02'

3rd-September, 2021

FORMATION OF DIFFERENTIAL EQUATION :

The differential equation can be formed by differentiating the ordinary equation and eliminating the arbitrary constant.

Example :

Form the Differential Equation by eliminating arbitrary constants in the following cases, and also write down the order of the differential equation.

$$\textcircled{1} - y = Ax + A^2.$$

$$y = Ax + A^2 \rightarrow \textcircled{1}$$

Differentiate w.r.t x we get, :

$$\frac{dy}{dx} = \frac{dy}{dx}(Ax) + \frac{dy}{dx}(A^2).$$

$$\frac{dy}{dx} = A + 0.$$

$$\frac{dy}{dx} = A. \quad \text{put value of } A \text{ in eq } \textcircled{1}.$$

$$\textcircled{1} \rightarrow y = \frac{dy}{dx} x + \left(\frac{dy}{dx}\right)^2$$

$$\frac{dy}{dx} \cdot x + \left(\frac{dy}{dx}\right)^2 - y = 0.$$

order = 1 , degree 2 .

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Exp :

Find the differential equation of the family of curves.

$$\textcircled{1} - y = Ae^{3x} + Be^{5x}$$

$$y = Ae^{3x} + Be^{5x} \rightarrow \textcircled{1}$$

Differentiate w.r.t x we get,

$$\frac{dy}{dx} = 3Ae^{3x} + 5Be^{5x} \rightarrow \textcircled{2}$$

Again Differentiate Eqn (2).

$$\frac{d^2y}{dx^2} = 9Ae^{3x} + 25Be^{5x} \rightarrow \textcircled{3}$$

Now, Multiplying Eqn 2 by 5 & subtracting Eqn 2 from Eq (3).

$$5 \frac{dy}{dx} = 5 (3Ae^{3x} + 5Be^{5x})$$

$$5 \frac{dy}{dx} = 15Ae^{3x} + 25Be^{5x}$$

$$+ \frac{d^2y}{dx^2} = + 9Ae^{3x} + 25Be^{5x}$$

~ subtract.

$$5 \frac{d^2y}{dx^2} - \frac{d^2y}{dx^2} = 6Ae^{3x}$$

$$\boxed{5 \frac{dy}{dx} - \frac{1}{6} \frac{d^2y}{dx^2} = Ae^{3x}}$$

 $\rightarrow \textcircled{A}$

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Now, Again
Eq 2 from
3

$$\textcircled{3} \rightarrow \frac{d^2y}{dx^2}$$

$$\textcircled{2} \rightarrow 3 \frac{dy}{dx}$$

$$\frac{d^2y}{dx^2} - 3 \frac{dy}{dx}$$

$$\boxed{10 \frac{d^2y}{dx^2} - 10 \frac{dy}{dx}}$$

putting eq
 $y =$

$$y = \frac{5}{6} \frac{dy}{dx}$$

$$y = \frac{dy}{dx}$$

$$y = \frac{dy}{dx}$$

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Now, Again Multiply 3 by eqn (2) and subtract eq 2 from 3.

$$3 \frac{dy}{dx} = 9Ae^{3x} + 15Be^{5x}.$$

$$\textcircled{3} \rightarrow \frac{d^2y}{dx^2} = 9Ae^{3x} + 15Be^{5x}$$

$$\textcircled{2} \rightarrow \frac{3 dy}{dx} = 9Ae^{3x} + 15Be^{5x}.$$

~ subtract.

$$\frac{d^2y}{dx^2} - \frac{3 dy}{dx} = 10Be^{5x}$$

$$\boxed{\frac{10 d^2y}{10 dx} - \frac{3 dy}{10 dx} = Be^{5x}.}$$

→ (B)

putting equation A & B in eq (1).

$$y = \frac{5 dy}{6 dx} - \frac{1}{6} \frac{d^2y}{dx^2} + \frac{10 d^2y}{10 dx} - \frac{3 dy}{10 dx}.$$

$$y = \frac{5 dy}{6 dx} - \frac{3 dy}{10 dx} - \frac{1}{6} \frac{d^2y}{dx^2} + \frac{10 d^2y}{10 dx}.$$

$$y = \frac{dy}{dx} \left(\frac{5}{6} - \frac{3}{10} \right) - \frac{d^2y}{dx^2} \left(\frac{1}{6} - \frac{10}{10} \right).$$

$$y = \frac{dy}{dx} \left(\frac{25 - 9}{30} \right) - \frac{d^2y}{dx^2} \left(\frac{5 - 3}{30} \right)$$

→

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$$y = \frac{dy}{dx} \left(\frac{8}{15} \right) - \frac{d^2y}{dx^2} \left(\frac{31}{15} \right)$$

$$y = \frac{dy}{dx} \left(\frac{8}{15} \right) - \frac{d^2y}{dx^2} \left(\frac{1}{15} \right)$$

~~$$\frac{8}{15} - \frac{1}{15}$$~~

$$\frac{8}{15} \frac{dy}{dx} - \frac{1}{15} \frac{d^2y}{dx^2} - y = 0$$

degree 1, order 2.

Q3

(3) ~~$y^2 = Ax^2 + Bx + C$~~

~~$y^2 = Ax^2 + Bx + C \quad \text{--- (1)}$~~

Differentiating Eq (1):

~~$\frac{d^2y}{dx^2} = 2Ax + B$~~

Differentiating Again

~~$\frac{dy}{dx} = 2A$~~

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H.W.

$$y^2 = Ax^2 + Bx + C.$$

$$y^2 = Ax^2 + Bx + C \quad \dots (1)$$

Differentiating w.r.t x :

$$\bullet \quad 2y \frac{dy}{dx} = 2Ax + B \quad \dots (2)$$

Again Differentiating w.r.t x :

$$2 \left[(y) \frac{d}{dx} \left(\frac{dy}{dx} \right) + \left(\frac{dy}{dx} \right) \frac{d}{dx} (y) \right] = 2A$$

$$2 \left[y \left(\frac{dy}{dx} \right)^2 + y \frac{d^2y}{dx^2} \right] = 2A \quad \dots (3)$$

Divide ² by Equation (3).

$$\frac{2}{2} \left[\left(\frac{dy}{dx} \right)^2 + y \frac{d^2y}{dx^2} \right] = \frac{2}{2} A$$

$$\left(\frac{dy}{dx} \right)^2 + y \frac{d^2y}{dx^2} = A \quad \dots (4)$$

Again Differentiating eq (4)

$$\frac{d}{dx} A = \frac{d}{dx} \left[\left(\frac{dy}{dx} \right)^2 + y \frac{d^2y}{dx^2} \right].$$

$$0 = \left[\frac{d}{dx} \left(\frac{dy}{dx} \right)^2 + \frac{d}{dx} \left(y \frac{d^2y}{dx^2} \right) \right]$$

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$$0 = \left[\left(\frac{d}{dx} \right) \frac{d}{dx} \left(\frac{dy}{dx} \right)^2 + \left(\frac{dy}{dx} \right)^2 \frac{d}{dx} \left(\frac{d}{dx} \right) \right] + \frac{d}{dx} \left(y \frac{d^2 y}{dx^2} \right)$$

$$0 = \left[\left(\frac{d}{dx} \right) \cdot 2 \left(\frac{dy}{dx} \right) \right]$$

$$0 = \frac{d}{dx} \left[\left(\frac{dy}{dx} \right)^2 + y \cdot \frac{d^2 y}{dx^2} \right]$$

$$0 = \frac{d}{dx} \left(\frac{dy}{dx} \right)^2 + \left[\left(y \right) \frac{d}{dx} \left(\frac{d^2 y}{dx^2} \right) + \left(\frac{d^2 y}{dx^2} \right) \frac{d}{dx} (y) \right]$$

$$= 2 \frac{dy}{dx} \left(\frac{d}{dx} \right) \left(\frac{dy}{dx} \right) + y \left(\frac{d^3 y}{dx^3} \right) + \frac{d^2 y}{dx^2} \frac{dy}{dx}$$

$$= 2 \frac{dy}{dx} \frac{d^2 y}{dx^2} + \frac{d^2 y}{dx^2} \frac{dy}{dx} + y \frac{d^3 y}{dx^3} = 0$$

$$\boxed{3 \frac{dy}{dx} \frac{d^2 y}{dx^2} + y \frac{d^3 y}{dx^3} = 0} \quad \text{Ans!}$$

Degree 1

order 3

(11)

(3) =

(4) - $y = A \cos x + B \sin x$

$$y = A \cos x + B \sin x \dots\dots (1)$$

Differentiate w.r.t x :

$$\frac{dy}{dx} = -A \sin x + B \cos x \dots\dots (2)$$

Again Differentiate Eqn (2)

$$\frac{d^2y}{dx^2} = -A \cos x - B \sin x$$

but we know that,

$$A \cos x + B \sin x = y$$

So,

$$\frac{d^2y}{dx^2} = -y$$

$$\frac{d^2y}{dx^2} + y = 0$$

$\therefore$ order = 2 (Two)

Degree = 1 (one).

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(5) $y = ae^x + be^{-x} + c \cos x + d \sin x$

$$y = ae^x + be^{-x} + c \cos x + d \sin x \dots (1)$$

Differentiating w.r.t x :

$$\frac{dy}{dx} = ae^x - be^{-x} - c \sin x + d \cos x \dots (2)$$

Again Differentiating w.r.t x

$$\frac{d^2y}{dx^2} = ae^x + be^{-x} - c \cos x - d \sin x \dots (3)$$

Taking 3rd Derivative

$$\frac{d^3y}{dx^3} = ae^x - be^{-x} + c \sin x - d \cos x \dots (4)$$

Taking 4th Derivative:

$$\frac{d^4y}{dx^4} = ae^x + be^{-x} + c \cos x + d \sin x \dots (5)$$

We know that,

$$ae^x + be^{-x} + c \cos x + d \sin x = y$$

So,

$$\frac{d^4y}{dx^4} = y$$

$$\frac{d^4y}{dx^4} - y = 0$$

order 4

degree 1.

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⑥ $y = A \cos 2t + B \sin 2t$.

→ ①.

$$y = A \cos 2t + B \sin 2t \text{ ----- ①}$$

Differentiation w.r.t t .

$$\frac{dy}{dt} = -2A \sin 2t + 2B \cos 2t \text{ --- ②}$$

Again differentiating w.r.t t ;

$$\frac{d^2y}{dt^2} = -4A \cos 2t - 4B \sin 2t$$

Take -4 as common;

$$\frac{d^2y}{dt^2} = -4 (A \cos 2t + B \sin 2t)$$

We know that, $(A \cos 2t + B \sin 2t) = y$.

$$\frac{d^2y}{dt^2} = -4 (y)$$

$$\frac{d^2y}{dt^2} + 4y = 0$$

Degree : 1

Order : 2.

arbitrary constant ki exact value nikalne key liye.

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Lecture 03¹⁷

10 September, 2024

Initial Value Condition:

Sometimes it is required to find the solution of a differential equation subjected to supplementary condition. If the conditions relate to one value of the independent variable such as $y = y_0$ at $x = x_0$ (written as $y(x_0) = y_0$) and $\frac{dy}{dx} = y'(x_0)$ at $x = x_0$ then

they are called initial value conditions, or (one point boundary condition) and x_0 is called the initial point.

An initial value problem consists of a D.E (of any order) together with a collection of initial value conditions that must be satisfied by the solution of the D.E and derivatives at the initial points

$$f(x) = y.$$

$$y = f(x)$$

$$y(x_0) = y_0$$

$$y'(x_0) = y_0$$

$$x = x_0$$

$$y = y_0.$$

example: $\frac{dy}{dx} = 2x \rightarrow (A)$

such that, $y(1) = 4$

$$x = 1, y = 4$$

This is an initial value problem, we note that

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10th Sep, 2026.

$$y = \text{?}$$

Integrating on Both sides of Eq. A.

$$\frac{dy}{dx} = 2x$$

$$\int dy = \int 2x dx$$

$$y = \frac{2x^2}{2} + C$$

$$y = x^2 + C \quad \dots \quad (1)$$

we note that,

 $y = x^2 + C$, C being arbitrary constant
in the general solution of (1).
Since, $y(1) = 4$.

Applying initial condition eq. (1) we get,

$$x = 1, \quad y = 4.$$

$$y = x^2 + C$$

$$4 = (1)^2 + C$$

$$4 = 1 + C$$

$$4 - 1 = C$$

$$C = 3$$

putting value of C in eq. (1).

$$y = x^2 + 3 \quad \text{Ans!}$$

just for understanding



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$$\frac{d^2y}{dx^2} + y = 0. \quad y(1) = 3, \quad y'(1) = -4$$

$x=1, y=3$ $x=-4$ and $y' = 1$

$$\frac{d^2y}{dx^2} + y = 0 \quad \dots \quad (1)$$

we know that,

$$\frac{d^2y}{dx^2} + y = A \sin x + B \cos x$$

So,

$$y = A \sin x + B \cos x \quad \dots \quad (2)$$

Now,

$$y' = A \cos x - B \sin x$$

Now, Apply initial condition $y(1) = 3$ in eq (1).

$$3 = A \sin(1) + B \cos(1)$$

Now, Apply initial condition $y'(1) = -4$ in eq (2).

$$-4 = A \cos(1) - B \sin(1)$$

Solving for A & B we get,

$$A = -4 \cos(1) + 3 \sin(1)$$

$$B = 3 \cos(1) + 4 \sin(1)$$

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$$-4$$

$$\text{and } y' = 1.$$

Boundary Value Conditions :

Definition :

The problem of finding the solution of a differential equation such that all the associated supplementary conditions relate to two different values of the independent variables. is called a two-point boundary value problem or simply a boundary problem.

The associated supplementary boundary conditions are called two-point boundary value conditions.

$$y(x_0) = y_0$$

$$y'(x_1) = y_1$$

Example :

Solve the boundary value problem:

eq (2).

$$\frac{d^2 y}{dx^2} + y = 0 \quad \dots (A)$$

$$y(0) = 1$$

$$x = 0, y = 1$$

$$y(\pi) = 5$$

$$x = \pi,$$

$$y = 5$$

Solution

Applying the boundary condition to the general solution.

$$y = A \sin x + B \cos x \quad \dots (B)$$

Apply given Boundary value in eq B.

$$x = 0, y = 1$$

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$$1 = A \sin(0) + B \cos(0)$$

$$1 = B(1)$$

$$B = 1$$

Now, Again put the given Boundary ^{condition} ~~equation~~ in equation B.

$$y(\pi) = 5.$$

$$\because \pi = 180^\circ$$

$$\sin(180) = 0$$

$$\cos(180) = -1$$

(B) $\rightarrow$

$$5 = A \sin \pi + B \cos \pi$$

$$5 = A \sin(180) + B \cos(180)$$

$$5 = A(0) + B(-1)$$

$$5 = -B$$

$$-B = 5 \quad \text{or} \quad B = -5$$

$$y(0) = 1 = B$$

$$y(\pi) = 5 = -B$$

Thus, we obtain 2 values of B and we are unable to determine any definite value of A.

Hence, the B-value problem has no solution.

Linear
Definition
An

F(x)

is so
function

A simil

It s

linear

(1) - The d
Degree

(2) - No

(3) - No
derivat

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10th September, 2026.

Linear Differential equations:

Definition

An ordinary D.E of the form

$$F\left(x, y, \frac{dy}{dx}, \dots, \frac{d^n y}{dx^n}\right) = 0.$$

is said to be linear if F is a linear function of the variable $x, \frac{dy}{dx}, \frac{d^2 y}{dx^2}, \dots, \frac{d^n y}{dx^n}.$

A similar definition applies to partial D.E.

It should be carefully noted that in a linear ordinary

- ① - The dependent variable y and its derivatives are all of Degree one.
- ② - No product of y and any of its derivatives appear.
- ③ - No transcendental function of y and its derivatives occur.

Exercise 9.1:

Q- classify each of the following equations as ordinary or partial differential equations: state the order and the degree of each equation and determine whether the equation is linear or non-linear.

$$(1) - \frac{d^3 y}{dx^3} + 4 \frac{d^2 y}{dx^2} - 5 \frac{dy}{dx} + 3x = \cos x$$

= ordinary D.E

order = 3

Degree = 1

= linear.

$$(2) - x^2 dy + y^2 dx = 0$$

$$\rightarrow x^2 dy = -y^2 dx$$

$$\frac{dy}{dx} = \frac{-y^2}{x^2}$$

= ordinary D.E

order = 3

Degree = 1

Linear

(3) -

(21)

$$(4) - \left(\frac{dy}{dx} \right)^2 = \left(\frac{d^2y}{dx^2} + y \right)^{3/2} = \text{ordinary D.E}$$

= order 2

= Degree 3

= Linear.

Multiplied 2 with the rising power:

$$\left[\left(\frac{dy}{dx} \right)^2 \right]^2 = \left(\frac{d^2y}{dx^2} + y \right)^{3/2 \times 2}$$

$$\frac{dy}{dx} = \left[\frac{d^2y}{dx^2} + y \right]^3$$

D.E

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10/sep

Q2 (1) - Form the equations of which the given function is a solution.

$$y = x + 3e^{-x} \rightarrow \text{ev (1)}$$

Diff w.r.t x we get

$$\frac{dy}{dx} = 1 + 3e^{-x} \frac{d}{dx}(-x)$$

$$\frac{dy}{dx} = 1 + 3e^{-x}(-1)$$

$$\frac{dy}{dx} = 1 - 3e^{-x} \rightarrow \text{ev (2)}$$

Add Both equations:

$$\frac{dy}{dx} + y = x + 1 + 3e^{-x} - 3e^{-x}$$

$$\frac{dy}{dx} + y = x + 1 \quad \text{Ans}$$

(11) $y = (x^3 + c) \cdot e^{-3x}$, c is arbitrary constant.
 $\rightarrow$ Differentiate using V.V formula.

$$\frac{d}{dx}(U \cdot V) = UV' + U'V$$

$$\frac{dy}{dx} = -3(x^3 + c)e^{-3x} + 3x^2 \cdot e^{-3x}$$

$$\frac{dy}{dx} = -3e^{-3x}(x^3 + c) + 3e^{-3x}x^2$$

$$\frac{dy}{dx} = 3e^{-3x}[-(x^3 + c) + x^2]$$

from ev (1) we get

$$y = (x^3 + c)e^{-3x}$$

$$\frac{y}{e^{-3x}} = x^3 + c$$

$$ye^{3x} = x^3 + c \rightarrow \text{put in eq (2)}$$

$$\frac{dy}{dx} = -3$$

$$\frac{dy}{dx}$$

$$\frac{dy}{dx}$$

Exercise 9:

Q1 -

Q2 -

Q3 -

Q4 -

Exercise

Q1 -

Initial

Q16

Exercise

Q1 -

Initial

10/sep

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the given

$$\frac{dy}{dx} = -3e^{-3x} \cdot e^{3x} y + 3e^{-3x} x^2$$

(base same power add)

$$\frac{dy}{dx} = -3y e^0 + 3e^{-3x} x^2$$

$\therefore e^0 = 1$

$$\frac{dy}{dx} = -3y(1) + 3e^{-3x} x^2$$

$$\frac{dy}{dx} = 3x^2 e^{-3x} - 3y$$

Exercise 9.1

H.W.

Important for
practice & paper.

~~Q1~~ ~~Q2~~ ~~Q3~~

Q1 - i - iv

Q2 - i - vii

Q3 - i - vi

Q4 - i - v

Exercise 9.2

Q1 - i - x

initial value problem =

Q16 - 19.

Do yourself.

Exercise 9.3

Q1 - 10. &

initial condition part 2 Q11-14 Do yourself.

Thursday

17th Sept 2025

Lecture 04

Q4. Solve the following initial value problems.

$$(i) \frac{dy}{dx} = -\frac{x}{y}, \quad y(3) = 4.$$

given that the D.E has $x^2 + y^2 = c^2$ as the general solution.

$$y = 3, \quad x = 4$$

$$x^2 + y^2 = c^2. \quad \text{--- (1)}$$

Now put the values of x and y in Equation 01.

$$(3)^2 + (4)^2 = c^2$$

$$9 + 16 = c^2$$

$$c^2 = 25$$

Taking Root On Both Side :

$$\sqrt{c^2} = \sqrt{25}$$

$$c = 5$$

put value of c in eq 1.

$$x^2 + y^2 = (5)^2.$$

$$y^2 = (5)^2 - x^2$$

Taking root On Both Sides :

$$\sqrt{y^2} = \sqrt{(5)^2 - x^2}$$

$$y = \sqrt{25 - x^2}$$

Remaining parts $\rightarrow$ Do yourself.

EQUATIONS OF THE FIRST ORDER AND FIRST DEGREE.

Solve the 1st order and 1st Degree differentiable equation by Separable Equations.

Definition:

A differential equation of the type

$$F(x) g(y) dx + f(x) g(y) dy = 0 \dots \textcircled{1}$$

is called an equation with variable separable or simply a separable equation.

(1) may be written as:

$$f(x) g(y) dx = -f(x) g(y) dy$$

$$\frac{f(x)}{f(x)} dx = \frac{-g(y)}{g(y)} dy$$

$$\int \frac{f(x)}{f(x)} dx + \frac{g(y)}{g(y)} dy = 0.$$

which can be easily integrate.

Working Rules:

Rule # 1 = Separate the variables.

$$f(y) dy = \phi(x) dx.$$

Rule # 2 =

Integrate both sides $\int f(y) dy = \int \phi(x) dx$

Rule # 3 =

Add an arbitrary constant (c) on R.H.S.

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Example 08:

Solve $\frac{dy}{dx} = \frac{x^2}{y}$

Solution:

Equations can be written as

Step # 1 = $y dy = x^2 dx$

Step # 2 = Integrate both sides.

$$\int y^2 dy = \int x^2 dx$$

$$\frac{y^3}{3} = \frac{x^3}{3} + C$$

Multiply both sides by 3.

$$y^3 = x^3 + C$$

$$3y^3 = 3x^3 + 3C$$

$$3C = C_2$$

$$\boxed{3y^3 = 3x^3 + C_2} \text{ Ans!}$$

Example 09.

$$\frac{dy}{dx} = \frac{1}{x \tan y}$$

Solution:

Equations can be written as:

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Step # 1 =

$$\tan y \, dy = \frac{1}{x} \, dx$$

Step # 2 : Integrate Both Sides =

$$\int \tan y \, dy = \int \frac{1}{x} \, dx$$

$$\rightarrow \ln \cos y = \ln x + c$$

$$\ln \cos y + \ln x = -c$$

$$\ln \cos y + \ln x + c = 0$$

$$\ln (x \cos y) = -c$$

$$\ln x \cos y = c$$

Applying exponential

$$(e)^{\ln |x \cos y|} = (e)^c$$

$$x \cos y = e^c$$

$$\rightarrow \ln a + \ln b = \ln a + b$$

$$x - c = c$$

(9)

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Example 10:

Solve

$$x \sin y dx + (x^2 + 1) \cos y dy = 0. \quad \text{--- (28)}$$

Solution:

$$x \sin y dx + (x^2 + 1) \cos y dy = 0 \quad \text{--- (1)}$$

Step # 01:

Separate the variables.

Divide by $(x^2 + 1) \sin y$, we get (1) as

$$\frac{x \sin y dx}{(x^2 + 1) \sin y} + \frac{(x^2 + 1) \cos y dy}{(x^2 + 1) \sin y} = 0$$

$$\frac{x}{(x^2 + 1)} dx + \frac{\cos y}{\sin y} dy.$$

Step # 02:

Integrate Both sides:

$$\int \frac{x}{x^2 + 1} dx + \int \frac{\cos y}{\sin y} dy = 0$$

$$\frac{1}{2} \int \frac{2x}{x^2 + 1} dx + \int \frac{\cos y}{\sin y} dy = 0.$$

$$\ln(x^2 + 1) + \ln \sin y = \ln c.$$

Multiplying 2 on Both sides:

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$$\ln(x^2+1) + 2 \ln(\sin y) = 2 \ln c.$$

$$\Rightarrow \ln x^n = n \ln x.$$

$$\ln(x^2+1) + \ln(\sin^2 y) = \ln c^2$$

$$\ln(x^2+1) \sin^2 y = \ln c$$

Taking Antilog on Both sides

$$(e)^{\ln(x^2+1) \sin^2 y} = e^{\ln c}$$

$$(x^2+1) \sin^2 y = c \quad \text{Ans!}$$

Example # 11:

Solve the initial value problem:

$$\frac{dy}{dx} = \frac{2x}{y+x^2y}, \quad y(0) = -2$$

$$x=0, y=-2$$

Solution:

$$\frac{dy}{dx} = \frac{2x}{y(1+x^2)}$$

Step # 1 =

$$y \, dy = \frac{2x}{1+x^2} \, dx$$

Step # 2 =

$$\int y \, dy = \int \frac{2x}{1+x^2} \, dx.$$

$$\frac{y^2}{2} = \ln(1+x^2) + C$$

Multiply Both sides by 2.

$$2 \frac{y^2}{2} = 2 \ln(1+x^2) + 2C$$

$$2C = C$$

$$y^2 = 2 \ln(1+x^2) + C$$

Taking Radical On Both sides:

$$y = \pm \sqrt{2 \ln(1+x^2) + C}$$

or

$$y = \pm [2 + \ln(1+x^2) + C]^{1/2}$$

Apply given condition:

$$x=0, y=-2.$$

$$-2 = \pm [2 \ln(1) + C]^{1/2}$$

~~$$-2 \neq \pm [2 \ln(1) + C]^{1/2}$$~~

$$-2 = \pm [2(0) + C]^{1/2}$$

$$-2 = \pm (C)^{1/2}$$

Squaring On Both sides

$$(-2)^2 = (C)^{1/2 \times 2}$$

$$\boxed{C = 4}$$

24th September

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